The ELIXIR Training Platform

ELIXIR All Hands 2022, 7-10 June, Amsterdam, Netherlands

The ELIXIR Training Platform

- Establishes and implements best practices in bioinformatics training through a training Toolkit (Training best practices, guidelines for trainers, standardized metrics for quality & impact assessment).
- Supports training providers across Europe in creating and delivering training for developers, researchers and trainers.
- Builds a sustainable training infrastructure.

http://www.elixir-europe.org/platforms/training

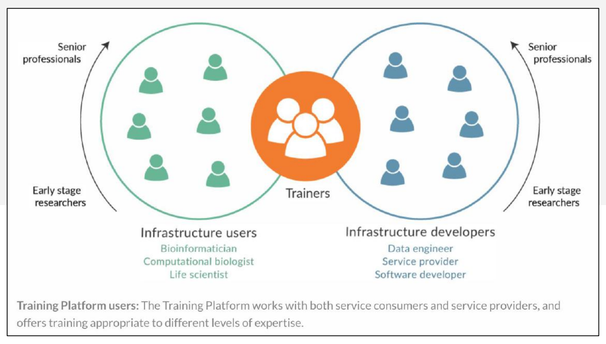
ELIXIR Training in Numbers (Training Metrics Database)

Since 2015	1,500+	38,700+	60+	4,700+
	training events	*people trained*	*countries*	*days of training*

ELIXIR Training - People infrastructure

- ★ **ExCos:** Fotis Psomopoulos (GR), Jessica Lindvall (SE), Celia van Gelder (NL, until 03/2022)
- ★ **Platform coordinator at the ELIXIR Hub**: Katharina Heil
- ★ **Task leaders**: 4 tasks, 16 co-leaders
- ★ **Working Groups:** 4 thematic Working Groups (eLearning, Certification, TtT, TeSS Club)
- ★ **Liaisons:** in Platforms, Communities, Focus Group and external projects
- ★ **Training Coordinators**: in the nodes
- ★ **Large pool of ELIXIR trainers** and **TtT trainers**
- ★ **All Nodes involved**

Target audiences

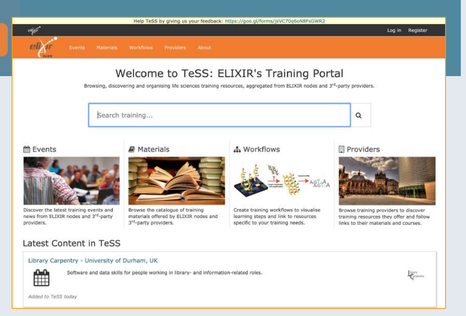


ELIXIR Training Platform Services

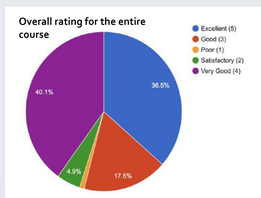
TeSS, the training registry for the ELIXIR community

- For trainers, the portal offers an environment for sharing materials and event information.
- For trainees, it offers a gateway to find relevant training events and resources, and to perform specific, guided analysis tasks via customised training workflows.
- For ELIXIR Nodes, TeSS provides opportunities to promote training events and news, and to contribute to a growing catalogue of materials.

Search TeSS (opens TeSS in a new tab): Search TeSS

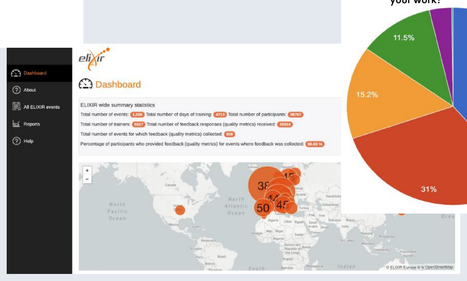


Statistics on May 2022:
- Registered training events: 12.684
- Registered training materials: 1.609
- Training content providers (organisations): 88



The Training Metrics Database (visit the database)

- For ELIXIR Training Coordinators, the database streamlines data collection, storage, and visualisation of ELIXIR Training quality and impact data.
- For anyone interested in ELIXIR training, it lets you browse and view ELIXIR-wide training summaries and interact with dynamic reports.



The ELIXIR-SI eLearning Platform EeLP (visit the platform)

- For trainers and trainees: EeLP provides a single access point to all necessary information, dynamic educational resources, tools and services for any training event, including access to registration, compute infrastructure (HPC), exams, comms and certificates.

Task 1
Training Toolkit

- **Publication and adoption of the ELIXIR Training Handbook**
  - Guidelines and best practices developed by the Training Platform.
  - Implemented search functionality in the ELIXIR resources for training organisers and trainers to improve discoverability of training publications.
  - Next goals: Handbook publication and new iterations for improvement, FAIRification of relevant documents and maintenance of the resource.
- **Toolkit expansion - impact assessment**
  - "Quality and Impact" ("Q&I") training assessment strategy measures:
    - **training quality** with short-term feedback immediately after the training event and
    - **training impact** with long-term feedback six months onwards after the event.
  - Next goal: increase the **"Q&I" strategy adoption** throughout ELIXIR
    - all ELIXIR Nodes are implementing the strategy;
    - six ELIXIR Nodes are implementing the impact assessment.

Task 2
Gap analysis, training materials development and training delivery

- **Objectives**:
  - Identify training gaps
  - Facilitate the development of FAIR training materials
  - Support ELIXIR Communities in developing Training Collections and Learning Paths
  - A template for ELIXIR certification of training material and courses
- **Planned short-term activities** (by end of 2022)
  - An iteration of the ELIXIR-wide **gap analysis survey**
  - A **lessons Hackathon** on FAIRifying training materials and Learning Paths
  - **Workshop on a roadmap** of the interaction between a **Community** and the Training Platform
  - Prepare a **Training collection** for **Software Best Practices**, together with the Tools Platform

Task 3
Consolidating quality and impact of the TeSS training resources

- **Quality, findability and interoperability of TeSS Training Resources**
  - Work towards a well defined annotation process of TeSS content
  - Enabling the adoption of Bioschemas by ELIXIR Nodes
- **Enabling ELIXIR Training Collections and Learning Paths in TeSS**
  - Training Collections active in TeSS - definition of best practises started
- **Assessing e-learning solutions across ELIXIR**
  - Review and curate e-learning material available across ELIXIR Nodes
  - Explore and expand the adoption of the ELIXIR-SI e-Learning Platform (EeLP) learning management system (LMS) across the nodes
- Co-production model between TrP Task 3 activities and the TeSS Infrastructure Service Roadmap
- Planned development of the ELIXIR Lesson Template and ELIXIR Training Courses Templates in EeLP to support collaborative training development and delivery

ELIXIR Training Platform Activities
within ELIXIR and beyond

- **TeSS Infrastructure Service**
  - Sets out to develop TeSS as a full fledged infrastructure service of production quality and widen its development base to multiple Nodes.

- **ELIXIR FAIR Training Focus Group**
  - Task Forces on; "Making Training FAIR" and "Training about FAIR"
  - Notable outcomes:
    PLOS CompBio "10 rules to make your training materials FAIR"
  - Currently working on:
    - Definition of FAIR Principles for Training Material
    - Lesson on FAIRification of Training Material
- **Generating training capacity in ELIXIR Communities and beyond**
  - Established liaisons between Communities and the TrP in order to identify common activities
  - Active participation to ELIXIR-relevant projects such as CONVERGE and EOSC-Life
  - Regular engagement to the Australia BioCommons
- **RITrainPlus**
  - Research infrastructures and core facilities exploring ways towards establishing a European School of Management for Research Infrastructures
  - ELIXIR (members) to contribute to course modules
- **ELIXIR in EOSC**
  - Co-chairs on Training in the ELIXIR EOSC Focus Group
  - Representation on the EOSC-A Advisory Group "Research careers and curricula"
    - Data stewardship curricula and career paths
    - Research careers, recognition and credit
    - Upskilling countries to engage in EOSC

Task 4
Training Capacity Building

- **Objectives**
  - **Train-the-Trainer (TtT):** Grow the skills of ELIXIR trainers as well as their number across Nodes & Create and support a community of ELIXIR trainers
  - **Supporting ELIXIR Management and operations:** Develop Node capacity in management and operations through specific actions
- **Achievements:**
  - **398** people trained since ELIXIR started over **32** courses (TtT)
  - Course design: Considerations for trainers – a Professional Guide (ELIXIR-GOBLET)
  - Created the ELIXIR TtT Instructor's community (~20 active members)
  - Developed training materials to deliver courses synchronously online
  - Survey to assess the pedagogical challenges and needs of trainers in DM/DS (TrP/Converge)
  - Supporting ELIXIR Management and operations
  - Assessment of needs in training of management and operations across Nodes
  - Node Coordinator & Data Manager Curriculum
- ***Planned activities ahead***
  - *Supporting the adoption of the ELIXIR Handbook of Operations*

Contact

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